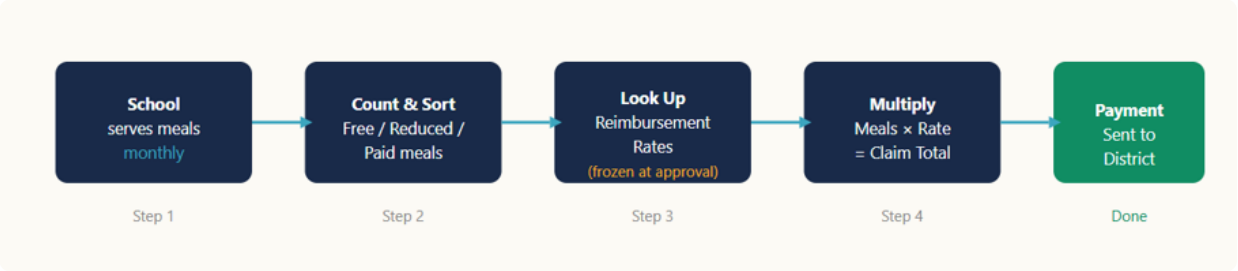
The Math

Basic Concept: Meals x Rate = Money

At its core, the equation is the number of qualifying meals served, multiplied by the government's reimbursement rate for each meal type, equals the dollar amount the federal government pays the school or program.

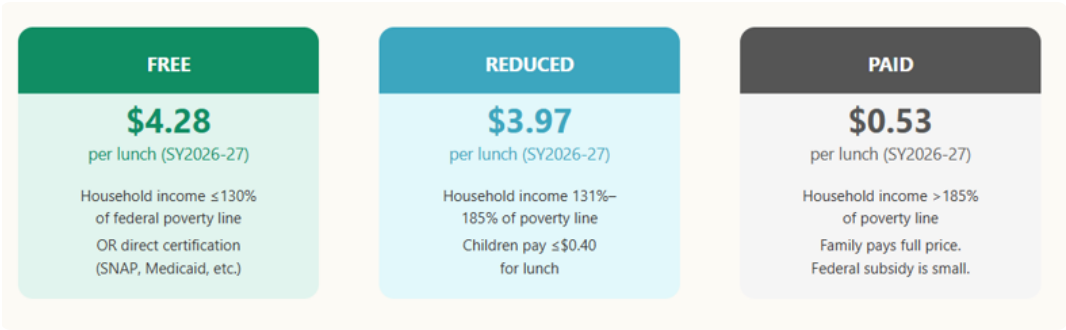
```
Meals Served  ×  Reimbursement Rate  =  Federal Payment
```



The federal government doesn't pay schools directly. USDA pays the *state agency* (like a state Department of Education), and the state pays the school districts. The math happens at every level of that chain.

Eligibility Categories

Not every meal is reimbursed at the same rate. USDA pays more for meals served to children from lower-income households. There are three categories, and each has its own rate.



The Full Rate Table (SY2026-27)

Meal Type	Free	Reduced	Paid	Notes
NSLP Lunch	$4.28	$3.97	$0.53	Plus state matching funds in some states

SBP Breakfast	$2.39	$2.09	$0.39	Severe Need schools get higher rate
NSLP Snack	$1.09	$0.55	$0.09	After-school programs only

Example:

Lincoln Elementary serves 420 lunches in October 2025:

- 295 Free meals × $4.28 = **$1,262.60**
- 52 Reduced meals × $3.97 = **$206.44**
- 73 Paid meals × $0.53 = **$38.69**

Total federal reimbursement for October: $1,507.73

How does the system know which children are Free vs. Reduced?

1. Families submit a household income application. A staff member reviews it and assigns the category.
2. The state runs an overnight match against SNAP and Medicaid records. Children who match are automatically certified Free without any application at all. This is called Direct Certification.

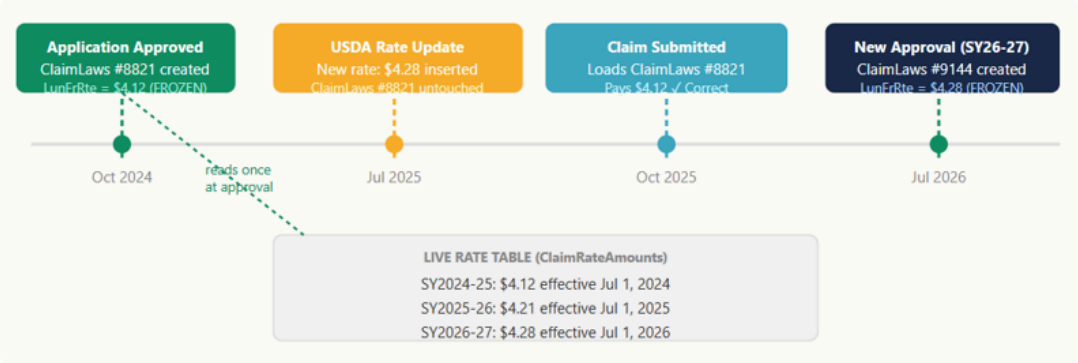
Locking in Rates

Here's a problem the system has to solve: USDA publishes new reimbursement rates every July 1. What happens to a claim that was submitted *before* July 1 but gets revised *after* July 1? Should it use the old rate or the new rate?

The answer is: **always the rate for the original claim month**. A revised claim for October 2024 processed in August 2025 uses the October 2024 rate — not the August 2025 rate. This is true even if the school year's rates have since been updated.

The system enforces this through a mechanism called **ClaimLaws**. When a school's application is approved, ClaimLaws records the school's **eligibility tier** — whether the school qualifies for the severe-need rate, which meal groups apply, and their operating parameters. Dollar amounts are **not** stored in ClaimLaws.

When a claim is computed, the system looks up the eligibility tier from ClaimLaws, then retrieves the corresponding dollar amounts from the rate table using the **original claim month** as the key. Because rates are indexed by date, a December 2024 claim always resolves to the December 2024 rate — even if it is revised or reprocessed after new rates have been published.



Before ClaimLaws existed, a rate update in July would silently change the reimbursement on *every prior-year revised claim that got reprocessed*. A school district disputing a claim from October 2024 would suddenly find their revised claim being recalculated at the July 2025 rate — which is wrong. ClaimLaws makes this structurally impossible.

What ClaimLaws stores: the eligibility tier (standard vs. severe need), the applicable meal groups, operating parameters, and the effective date range. This is the school-specific determination made at application approval time. The dollar amounts that correspond to that tier for any given claim month are looked up separately at compute time — keeping rate updates a simple data operation that never touches approved applications.

Community Eligibility Provision (CEP)

Collecting individual household income applications is a lot of work — for families, for school staff, and for state agencies. In high-poverty schools where almost everyone qualifies anyway, it creates paperwork for little benefit. The Community Eligibility Provision (CEP) solves this by letting qualifying schools skip individual applications entirely and serve *all students* free meals.

But the federal government still needs to know what to pay. They can't just pay the free meal rate for every meal — not every student at a CEP school is actually from a low-income household. So USDA created a formula that estimates the right reimbursement based on how many students are *directly certified* (automatically verified through SNAP, Medicaid, etc.) without requiring individual applications.

The ISP Formula

```
ISP = Identified Students ÷ Total Enrollment × 100
```

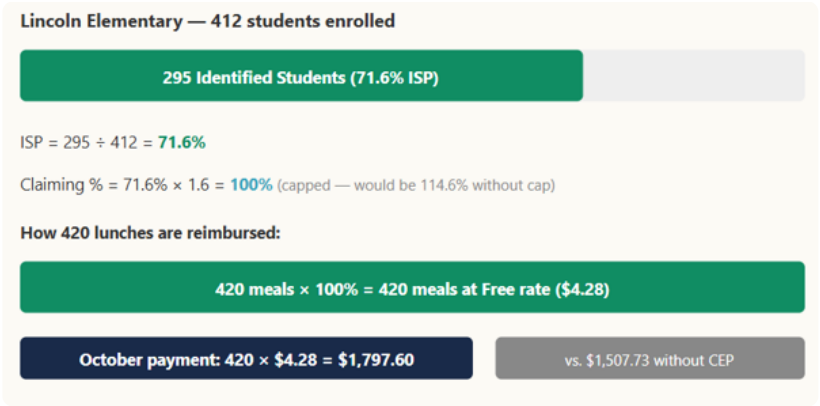
The **Identified Student Percentage (ISP)** is the share of enrolled students who are directly certified through a government benefit program. A school with 300 students where 180 are directly certified has an ISP of 60%.

To qualify for CEP, the ISP must be at least 40%.

The Claiming Percentage Formula

```
Claiming % = ISP × 1.6  (capped at 100%)
```

USDA multiplies the ISP by 1.6 to produce the **Claiming Percentage** — the share of all meals reimbursed at the free rate. The multiplier of 1.6 is set by federal regulation and cannot be changed.



CEP Groups

A school district doesn't have to elect CEP school-by-school. They can pool schools into a **CEP Group**, combining all those schools' enrollment and identified students into one ISP calculation. This helps schools that are just under the 40% threshold by averaging them with higher-qualifying schools in the same group.

Example

Salt Lake City SD has three west-side schools. Individually, two qualify but one doesn't:

- Lincoln Elementary: 295/412 students identified = 71.6% ISP ✓
- Roosevelt Elementary: 180/380 students identified = 47.4% ISP ✓
- Jefferson Elementary: 140/410 students identified = 34.1% ISP ✗ (below 40%)

Pooled as a group: 615 identified ÷ 1,202 enrolled = **51.2% ISP → 81.9% Claiming %**. All three schools join CEP together.

The 1.6 multiplier is federal law. It cannot be configured The system must not allow any state-specific override of this multiplier. It is set in 42 U.S.C. § 1759a(a)(1)(F)(viii). Any attempt to make it configurable would constitute a federal compliance violation.

Provision 2 & 3: Schools That Don't Count Applications Every Year

Before CEP existed, USDA offered two older programs for high-poverty schools to reduce paperwork. Both are still in use today.

Provision 2

Under Provision 2, a school establishes a **base year** by collecting full household income applications once, measuring the percentage of Free/Reduced/Paid meals accurately. For the next three years (the non-base years), the school claims using those same percentages — they don't collect new applications or recount eligibility. They just report total meals served and the system applies the base-year percentages.



Provision 2 Math

Base year established: 68% Free, 12% Reduced, 20% Paid.

In Year 3, Provo serves 22,840 total lunches in September:

- 22,840 × 68% = 15,531 Free meals × $4.28 = **$66,472.68**
- 22,840 × 12% = 2,741 Reduced meals × $3.97 = **$10,881.77**
- 22,840 × 20% = 4,568 Paid meals × $0.53 = **$2,421.04**

Total: $79,775.49 — calculated from base-year percentages, not individual eligibility records.

Provision 3

Provision 3 also eliminates annual application collection, but its mechanism is different from Provision 2. Rather than simply applying base-year percentages to reported meal counts, Provision 3 captures richer base-year data — including **claiming percentages** (share of total meals that were Free, Reduced, and Paid) and **eligibility percentages** (share of enrolled

students in each category) — and uses those ratios to project non-base year meals from enrollment and operating days alone.

In a non-base year, the sponsor enters only two values: **current enrollment** and **number of operating days**. The system then:

1. Scales the base-year total meal count proportionally for enrollment change and operating days
2. Applies base-year claiming percentages to get the Free/Reduced/Paid breakdown
3. Feeds those counts into the **standard reimbursement calculation at current USDA rates**

There is no separate inflation adjustment — current rates apply automatically through the normal annual rate update process, the same as any other claim. The goal is administrative simplification: eliminating annual application collection at high-poverty schools where eligibility is high and stable. Far less common than Provision 2.

When a Provision 2 or 3 school is found to have overclaimed, the state agency can't simply compare meal counts against individual eligibility records because those records don't exist for non-base years. USDA requires a different fiscal action formula (FA2 or FA3) specifically for Provision schools. This is why the compliance system has four separate fiscal workbook variants.

CACFP: Daycare and After-School Programs

Child and Adult Care Food Program (CACFP) works on the same Meals × Rate = Money principle as school programs, but the rate structure is different. In CACFP, the rate you receive depends on the **type of provider**, not just the child's income level.

Two Tiers of Providers

CACFP day care homes (individual home-based childcare providers) are divided into two tiers based on the provider's own income or the income level of the neighborhood where they operate.



Meal Patterns Are Different

CACFP also has stricter meal pattern requirements than school programs. A CACFP breakfast must include a grain, a milk, and a fruit or vegetable. A lunch must include a meat/meat alternate, a grain, two different vegetables or fruits, and milk. The system checks that claimed meals meet the pattern — meals that don't meet the pattern are disallowed even if the count is otherwise correct.

Meal Limits Per Day

CACFP day care homes can claim a maximum of **two meals and one snack** per child per day, or **one meal and two snacks** per child per day. A child cannot be claimed for three full meals. This cap is enforced in the claim validation logic.

Centers vs. Homes

CACFP covers both institutional centers (a licensed daycare facility) and individual home-based providers (someone running daycare from their house). Centers claim at a single rate based on enrolled children's individual eligibility. Homes use the Tier I/II system. The claims data model handles these separately.

SFSP & SSFO

Summer meal programs are designed to be simpler than school-year programs. Instead of tracking individual children's eligibility, summer programs operate as **open sites** where any child 18 or under who shows up eats for free. No applications. No eligibility sorting. No Free/Reduced/Paid categories.

Because all meals are free, there's only one reimbursement rate per meal type.



The Attendance Requirement

Summer programs can only claim meals for children who are actually present. Unlike schools that can project meal counts, SFSP sites must record actual attendance. The system enforces a **50% attendance cap** at some site types — you can't claim more meals than half the enrolled capacity if actual attendance records aren't available.

Non-Congregate Meals (SFSP)

Traditional summer meal rules require children to eat together at the site — called congregate service. After COVID, USDA permanently allowed some sites to offer **non-congregate** meals: grab-and-go bags, parent pick-up, or delivery to children who can't reach the site. Non-congregate service has stricter recordkeeping requirements and the system enforces three distinct modes:

- **Congregate only** — children eat at the site, standard rules apply
- **Non-congregate only** — all meals are distributed for off-site consumption
- **Mixed** — some meals eaten on-site, some distributed; each must be tracked separately

SSFO — Seamless Summer Food Option

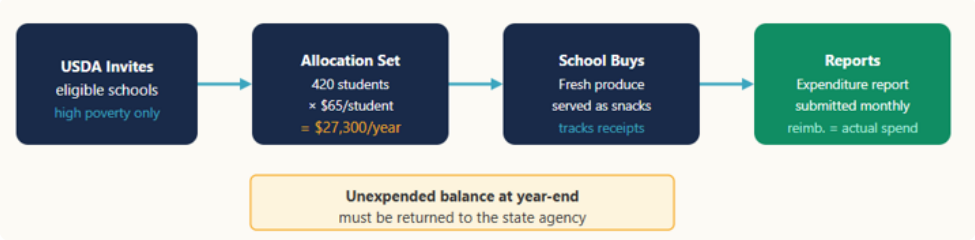
Some schools want to keep serving meals through summer without switching from their NSLP paperwork to SFSP paperwork. The Seamless Summer Food Option (SSFO) lets them operate under school-year administrative rules but with summer funding. Their reimbursement rates are similar to NSLP but come from a different federal grant account. The system keeps SSFO claims separate from both NSLP and SFSP claims to prevent the different rules from mixing.

Why SFSP rates are higher per meal than NSLP free ratesSFSP is fully federally funded — states and local programs receive no additional state funds or family copayments. The higher per-meal rate compensates for the lack of other funding sources and the higher administrative cost of summer-only operations.

FFVP: Allocation Math

The USDA gives each selected school a fixed dollar amount for the year based on their enrollment, and the school spends that money on fresh produce served as snacks. USDA sets the per-student amount each school year (approximately $50–$75 per student). Only high-poverty elementary schools are eligible, typically those with 50%+ free and reduced eligibility, and selection is by state agency invitation, not application.

```
Annual Allocation = Enrolled Students × Per-Student Amount
```

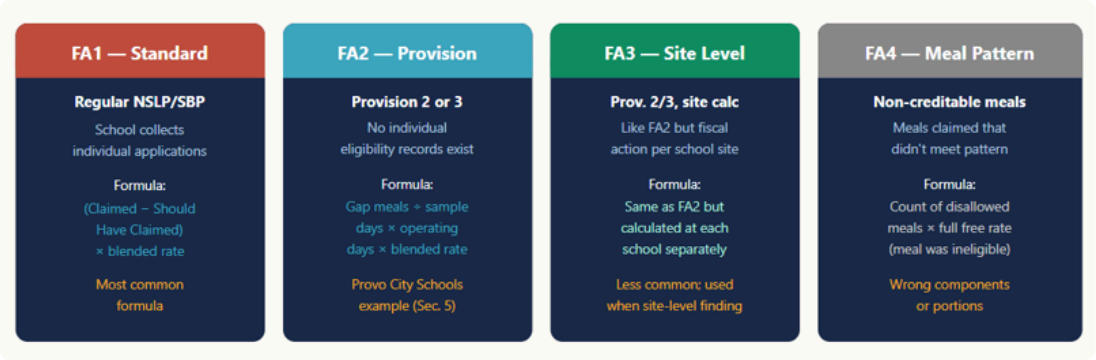


No ClaimLaws needed for FFVP Because FFVP reimbursement equals actual expenditure rather than meals × rate, there's no reimbursement rate to freeze. The school is simply reimbursed for what they spent, up to their allocation cap.

Fiscal Action Math

When a state agency reviews a school or program and finds that they claimed more meals than they actually served or claimed meals that didn't meet the program requirements the state must calculate how much money needs to be paid back. This is called a **fiscal action**.

USDA requires specific formulas depending on what type of school is being reviewed. There are four variants, called FA1 through FA4.



The Five-Day Reconciliation (The Most Common Finding Source)

During an on-site review, the reviewer doesn't examine every single day of a school year. Instead, they pick a sample — usually five days from a recent month's claim. They compare the number of meals the kitchen *produced* on each of those days against the number the school *claimed* on each of those days. If there's a gap, they extrapolate it across the full year.

Five-Day Reconciliation + FA2 — Provo City Schools (Provision 2 School)

Reviewer samples 5 days from September 2025. Total gap found: **312 meals over 5 days**.

September had 21 operating days. FA2 formula:

```
312 meals ÷ 5 sample days × 21 operating days = 1,310 estimated overclaimed meals
1,310 × $4.05 blended rate = $5,305.50 assessed repayment
```

The $5,305.50 moves to the Accounting domain as a repayment schedule. Provo's next disbursement is reduced by that amount.

The *calculation* of how much is owed is an enforcement function living with the review record.

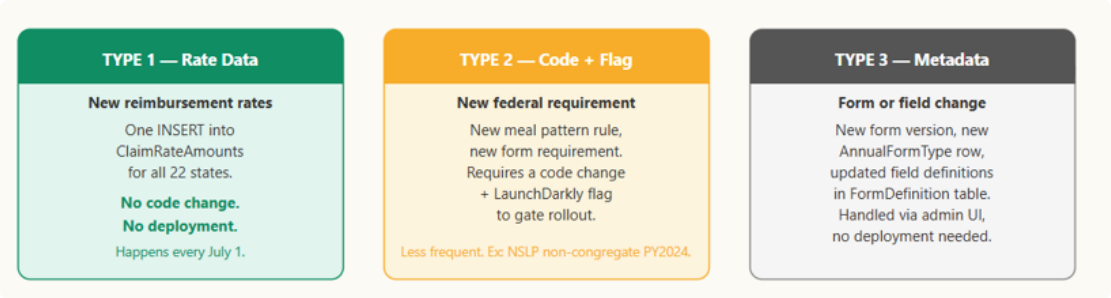
The *collection* of what's owed (the repayment schedule, the check received) lives in Accounting. Accounting only receives the final dollar number. It never needs to know whether

FA1, FA2, FA3, or FA4 was used.

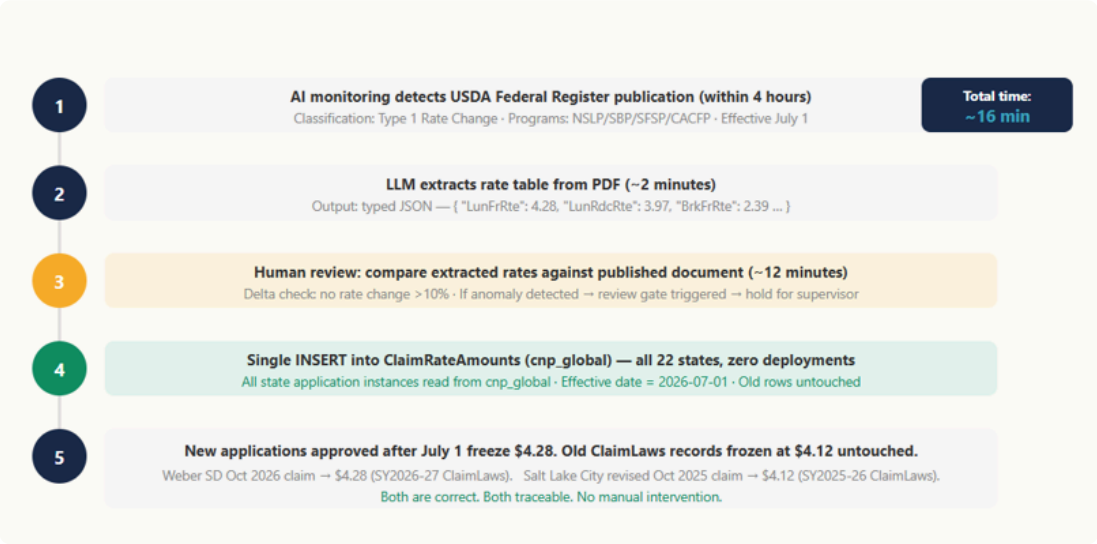
Annual Rate Updates:

Every July 1, USDA publishes new reimbursement rates for the upcoming school year. In the old architecture, applying these new rates meant updating stored procedures, redeploying code to each state, and hoping nothing broke. In the new architecture, it's a single database INSERT that takes effect for all states simultaneously — no code changes, no deployments.

Three Types of Annual Federal Changes



The Type 1 Rate Update in Detail



In DIS and CTS, reimbursement rates are often embedded in stored procedure logic, not in a table. A rate change means finding every SP that referenced the old rate, updating each one, testing, and deploying per state. The new design makes rates data, not code.

Anomaly Gate

USDA occasionally publishes corrections and has issued erroneous rate documents before. The system flags any rate increase or decrease greater than 10% from the prior year for mandatory human review before the INSERT is allowed. A 3–4% annual increase is normal (inflation

adjustment). A 15% jump would be suspicious and must be confirmed against the official Federal Register document before going in.